Machine Learning – Modeling and Learning

15-110 – Monday 4/13

Learning Goals

- Given a dataset, identify **features** which may help predict information about the data
- Identify **common types** of machine learning algorithms

Machine Learning Overview

Machine Learning Finds Patterns

Machine Learning is the process of identifying patterns in data to build a **model**. These models can then be applied to new data, to gain insights or make predictions about it.

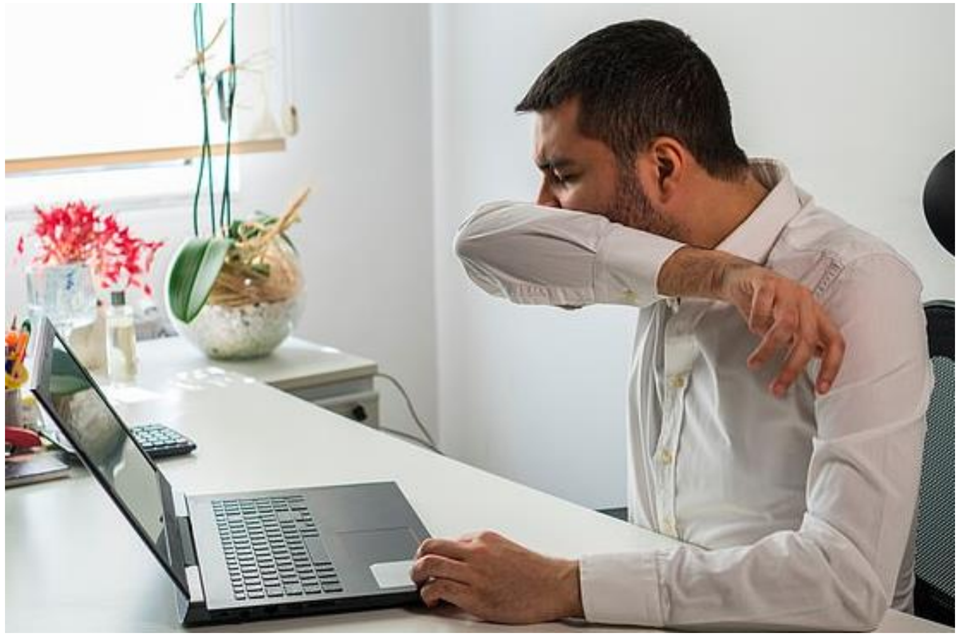
Machine Learning algorithms specialize in **optimizing their own performance** at some task by gaining experience (processing new data).

It's like human learning – you need to **practice** and get **feedback** to learn how to do a new task.

Machine Learning and COVID-19

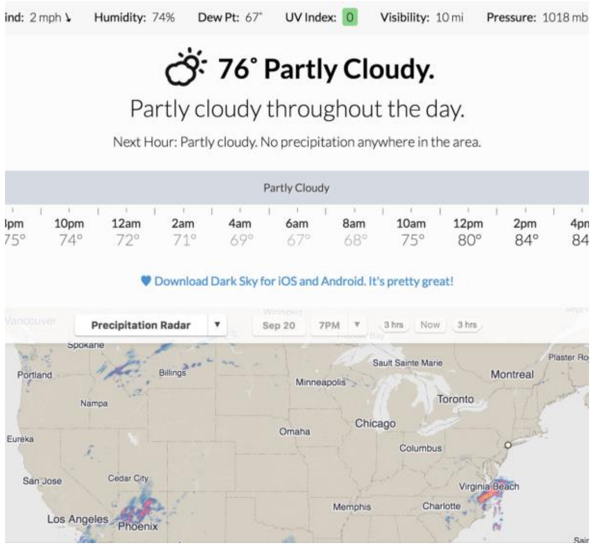
Computer scientists can develop machine learning algorithms that help identify who might have a virus as early as possible.

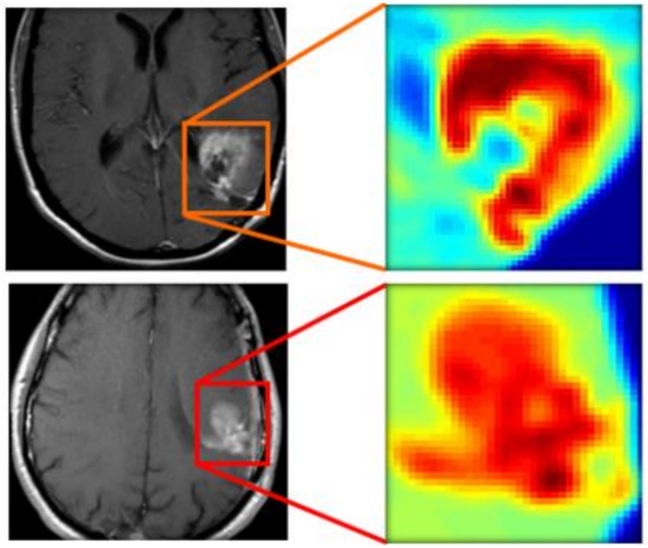
For example, CMU researchers are working on an algorithm that can detect if someone has COVID-19 by listening to them cough.



Machine Learning in Many Contexts

Machine Learning is used in hundreds of contexts across the world, including speech recognition, weather prediction, and medical diagnosis.





Machine Learning Main Types

Machine Learning algorithms are divided into two main types. Which type you use depends on whether the data you have has been **labeled** with the answers you're looking for.

Supervised: Used when the data is labeled. The goal is to learn to predict the best output given unlabeled data, by **training** on the labeled data.

Unsupervised: Used when the data is **not** labeled. The goal is to infer the natural structure of the data.

ML Process: Choose, Train, then Test

To apply machine learning to a supervised task, you can follow a simple process.

First: **choose** which algorithm you want to use and which features you'll train on. There are many different kinds of machine learning algorithms; we'll discuss some common ones soon.

Second: **train** the model created by the algorithm by providing it with data. The algorithm will 'learn' from the data the same way a student learns by going over worked examples.

Third: **test** the model on a different set of data. This helps determine how accurate the model actually is.

Training Identifies Key Features

To use machine learning, we must break down a complex data entry into a collection of **features**. During the training process, the algorithm identifies which features contribute the most to the underlying pattern.

If you have a table of data, the features would be the **columns**; in our ice cream data, the three features would be the three chosen flavors.

You can create and add new features, the same way you would add new columns in data analysis.

Three Types of Data

When we work with simple table data (in data analysis or machine learning), that data often falls into one of three types.

Categorical: Data fall into one of several categories. Those categories are separate and cannot be compared.

Example: type of fruit (apples, bananas, oranges)

Ordinal: Data fall into separate categories, but those categories **can** be compared – they have a specific **order**.

Example: how good you are at cooking (bad, okay, good, very good, excellent)

Numerical: Data are **numbers**. We can perform mathematical operations on it and compare it to other data.

Example: temperature (50° F vs 72° F)

Example: Is It a Dog?

???

Dog Features:

- Ear type = pointy
- Has Fur = True
- Screen in background = True

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Example: Is It a Dog?

Dog Features:

- Ear type = pointy
- Has Fur = True
- Nose length < 6in

???

Feature Takeaways

It's rare for a machine learning algorithm to identify a single feature that can definitively be used to answer a question.

Usually, the algorithm uses a **combination of several features**, which are weighted based on how well they correlate with the correct answer.

The algorithm needs to learn from a **lot** of examples to get a good sense of what the real pattern is.

Learning Advanced Features

What about text data?

The features might break down the words in the text in different ways – whether the word occurs in an entry, how often words occur, which two-word and three-word phrases occur, etc. This is commonly referred to as **natural language processing (NLP)**.

What about image data?

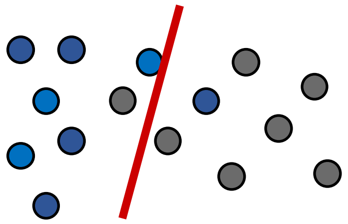
Identifying features in images is hard. The computer may analyze an image to identify different colors, the image's depth, where lines occur, and more. This is commonly referred to as **image recognition**.

Ted Talk: How we teach computers to understand pictures

Common Types of ML Algorithms

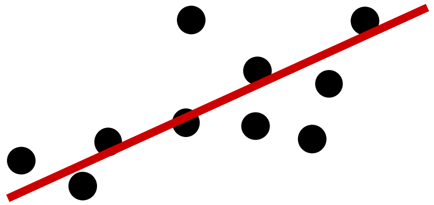
Machine Learning Algorithm Types

There are many machine learning algorithms, but a lot of them can be grouped into one of three types.



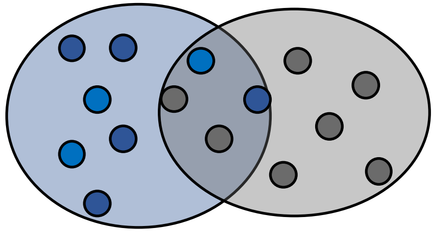
Classification – what **category** does a data entry belong to?

Uses **labeled** data



Regression – what **score** should a data entry be assigned?

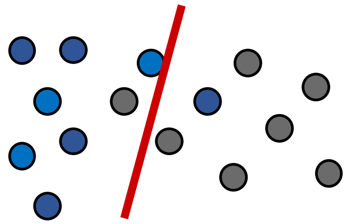
Uses **labeled** data



Clustering – how do we break a set of data entries into **groups**?

Uses **unlabeled** data

Classification Algorithms



A **classification** algorithm takes data of any type and produces a **model** (called a classifier) which determines which **category** that set of data most likely belongs to. In other words, it produces a **categorical** (or **discrete**) result.

For example, a classifier might take in data about a student (their classes, organizations they're a part of, their year) and use it to predict their major.

Classification is commonly used in spam filters and image recognition.

Example: Naive Bayes Classifier

Naive Bayes is a simple classification algorithm. The classifier it creates uses **conditional probabilities** to predict categories.

When Naive Bayes is given a new data entry, it uses the data it has already collected to determine the **probability** that that data entry belongs to each possible category.

$$P(\mathrm{A}|\mathrm{B}) = \frac{P(\mathrm{B}|A) \cdot P(A)}{P(B)}$$

For each category C, it checks how **probable** it is that each feature occurs in C, by checking how often it occurred in past data entries. By combining these probabilities together, it can form a general probability that the data entry belongs in that category.

The category with the highest probability wins!

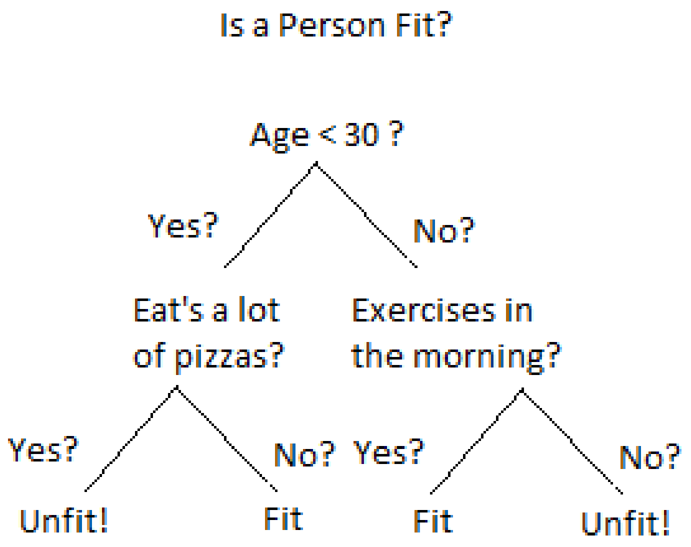
Example: Decision Tree Classifier

Decision Trees are classification algorithms that create models which choose categories using **trees**.

Each inner node of the tree is a question about some feature(s) in the data entry. The answer leads to one of the possible children.

The leaves of the tree are then the categories that are chosen.

The tree is designed to maximize the number of previous data entries that are mapped to the correct category.

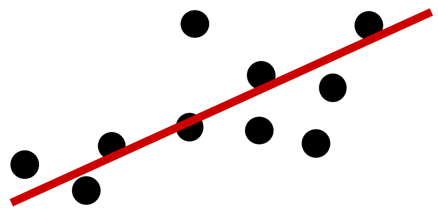


Other Classification Algorithms

Other classification algorithms include:

- **Logistic Regression** – determine whether a data entry falls into one of two categories
- **K-Nearest Neighbors** – find the categories of the K data entries closest to the new data point, and choose the most common one. (Also used in regression).
- **Neural Networks** – use the initial features to repeatedly create new 'hidden' features, and use those to classify data. (Also used in regression).

Regression Algorithms



A **regression** algorithm takes numerically-labeled data and produces a function that maps data entries to **numerical** (or **continuous**) results. Instead of predicting a specific category, it gives each data entry a **score**.

For example, a regression algorithm might take data about an incoming student (their high school GPA, AP scores, SAT score, etc.) and use it to predict their first-semester GPA.

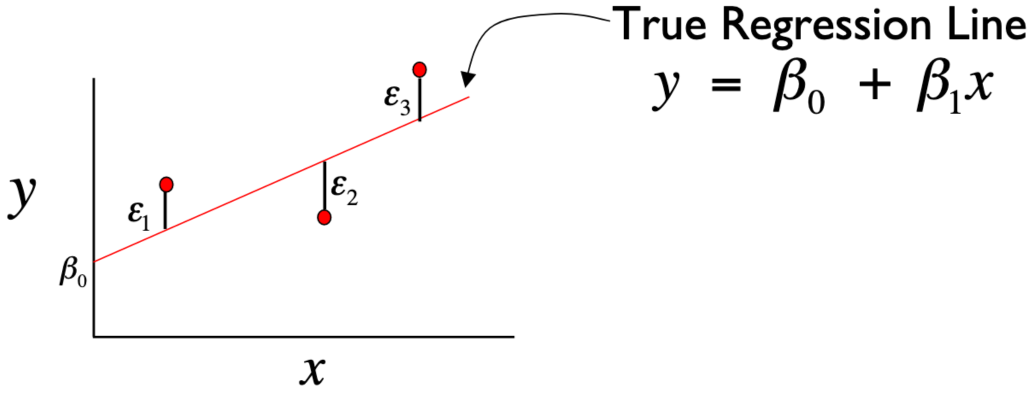
Regression algorithms are used to predict things like stock market prices and weather temperatures.

Example: Linear Regression

As a reminder, a **regression line** is a line produced by a statistical analysis that 'fits' a data set.

A **linear regression** algorithm produces a regression line through entries that minimizes the distance between a data entry and the line.

This can become complicated when there are many different features that all contribute to the algorithm's result.



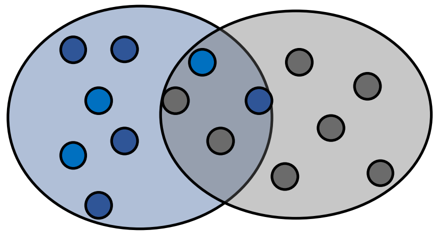
Regression Cautions

Regression algorithms can be very powerful, but you have to be careful with them.

If there are **more features that data entries**, you won't be able to solve for the equation.

If there is **non-constant error** in the actual results, the regression will do a poor job modeling it.

Clustering Algorithms



A **clustering** algorithm takes a set of data and produces a model that breaks the data into some number of **clusters**, by identifying data entries that are similar to each other.

Clustering is an **unsupervised learning algorithm**; therefore, we don't actually know the true categories or scores of the provided data. This makes it hard to 'test' the resulting model, as we aren't sure what the actual result should look like.

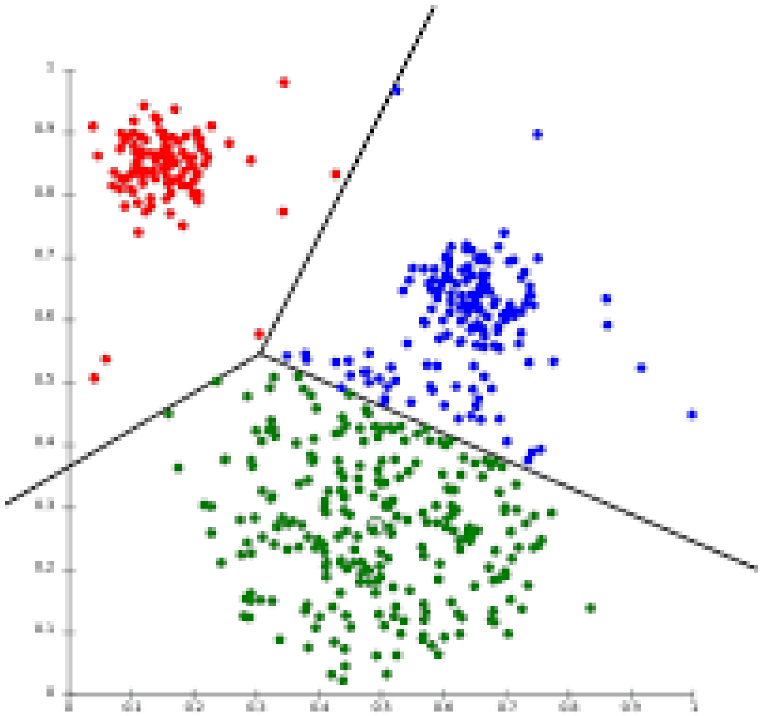
Clustering is used in data compression and in bioinformatics.

Example: K-Means Clustering

The **k-means clustering** algorithm takes a set of data and generates a model that breaks it into k different clusters.

Data points are assigned to the cluster that has a center closest to its **mean,** which is computed from its features. The algorithm tries to place clusters so that each cluster has as small a radius as possible.

Once an algorithm has clustered data, a human has to go in to determine if the clusters have common patterns that can provide meaning to the grouping.



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